Projection

$\mathrm{Proj}_S(x_0) = \arg\min \|x - x_0\|_2^2 \quad \text{s.t.} \quad x \in S$ 正交投影

可理解为 $I_S(x)$ 的 proximal

$\mathrm{prox}_f(x) = \arg\min \frac{1}{2}\|y - x\|_2^2 + I_C(x)$

$= \underset{y \in S}{\arg\min} \frac{1}{2}\|y - x\|_2^2 \quad := \mathrm{Proj}_S(x)$

可理解为梯度投影，一步就收敛

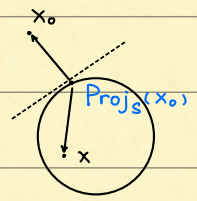
$\nabla f^k = 2(x^k - x_0), \quad \alpha = \frac{1}{L} = \frac{1}{2}$

$x^{k+1} = \mathrm{Proj}_S(x^k - (x^k - x_0)) = \mathrm{Proj}_S(x_0)$

1st Proj thm：凸集投影必唯一

2nd Proj thm：凸集投影有

$\langle P_S(x_0) - x_0, \ P_S(x_0) - x \rangle \le 0, \quad \forall x \in S$



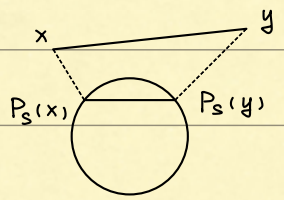
Pf：当 f 凸且 S 凸

$\langle \nabla f, x - x^* \rangle \ge 0, \ \forall x \in S$

$f = \frac{1}{2}\|y - x\|_2^2, \quad \nabla f = x^* - x_0$

=> Nonexpansiveness 非膨胀性

$\|P_S(x) - P_S(y)\| \le \|x - y\|$



Pf：把任意的 x 代入为 $P_S(y)$.

$\langle P_S(x) - x, \ P_S(x) - P_S(y) \rangle \le 0$

反着再代一次，相加

$\|P_S(x) - P_S(y)\|^2 \le \langle P_S(x) - P_S(y), \ x - y \rangle$

$\le \|P_S(x) - P_S(y)\| \cdot \|x - y\|$

[例]

Non-Negative $S = \mathbb{R}^+$, $\mathrm{Proj}_S(x) = \mathrm{ReLU}(x)$

Affine $S = \{x \mid y = Ax\}$ $\quad \mathrm{Proj}_S(x) = x - A^H(AA^H)^{-1}(Ax - y)$

$$= \mathrm{Proj}_A^{\perp}(x) + A^{\dagger}y$$

Sparse $S = \{x \mid \|x\|_0 \le K\}$. $\mathrm{Proj}_S(x) = H_K(x)$

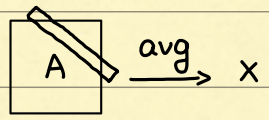
Low Rank $S = \{A \mid \mathrm{rank}(A) \le r\}$. $\mathrm{Proj}_S(A) = \mathrm{tSVD}_r(A)$

PSD Symmetric $\mathrm{Proj}_{S^+}(A) = [\frac{A + A^H}{2}]_+ = [\mathrm{Sym}(A)]_+$

Linear Structure

$$\min \|A - X\| \quad \text{s.t. } X \in \mathcal{L} = \{X = \sum_{i=1}^{s} x_i \underbrace{u_i}_{\text{Basis}}\}$$

e.g. Toeplitz $u_1 = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \cdots u_s = \begin{bmatrix} & & 1 \\ & \ddots & \\ 1 & & \end{bmatrix}$



Alternative Projection 交替投影

$\min \|A - X\|$ s.t. $X \in S_1 \cap S_2$ $\qquad X^{k+1} = P_{S_2}(P_{S_1}(X^k))$

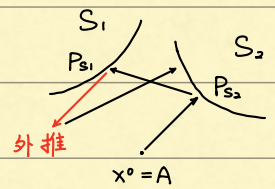
e.g. 非负方程 $y = Ax$, $x \ge 0$

$$x^{k+1} = [x^k - A^H(AA^H)^{-1}(Ax - y)]_+$$

Dykstra

$Y^{k+1} = \mathrm{Proj}_{S_1}(X^k - Z^k)$ $\qquad X^{k+1} = \mathrm{Proj}_{S_2}(Y^{k+1} - W^k)$

$Z^{k+1} = Z^k - Y^{k+1} + X^k$ $\qquad W^{k+1} = W^k - X^{k+1} + Y^{k+1}$



S_1, S_2 convex 必收敛

ADMM

$\min \|A - X\|$ s.t. $X \in S_1 \cap S_2$

$\min \|A - X\|^2 + I_{S_1}(X) + I_{S_2}(Y)$ s.t. $X = Y$

$$AL = \|A - X\|^2 + I_{S_1}(X) + I_{S_2}(Y) + \langle Z, X - Y\rangle + \frac{1}{2}\rho\|X - Y\|^2$$

$$= \|A - X\|^2 + I_{S_1}(X) + I_{S_2}(Y) + \frac{1}{2}\rho\|X - Y + \frac{1}{\rho}Z\|^2 - \frac{1}{2\rho}\|Z\|^2$$

$$X^{k+1} = \mathrm{argmin}\ \|A - X\|^2 + \frac{1}{2}\rho\|X - Y + \frac{1}{\rho}Z\|^2 \quad \text{s.t. } X \in S_1$$

$$= \mathrm{Proj}_{S_1}(\frac{2}{2+\rho}(A + \frac{1}{2}\rho Y^k - \frac{1}{2}Z^k))$$

$$Y^{k+1} = \operatorname{argmin} \| X - Y + \frac{1}{\rho} Z \|^2 \quad \text{s.t. } Y \in S_2$$

$$= \operatorname{Proj}_{S_2}\left(X^{k+1} + \frac{1}{\rho} Z^k \right)$$

$$Z^{k+1} = Z^k + X^{k+1} - Y^{k+1}$$

当 $\rho = 1$ 时，且 S_1 为线性空间，ADMM 即 Dykstra (?)